

A Rational Quantum Channel Beyond Finite Tracial Baths

A construction on fifteen system qubits

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October 2026 (version 1.0)

Abstract

A quantum channel on a finite system always has a finite realization with a pure environment. With a maximally mixed environment this can fail even approximately: since $\text{MIP}^* = \text{RE}$, some factorizable channels lie outside the closure of the finite matrix-tracial factorizations. We construct an explicit one: a unital channel on fifteen qubits with 11 230 Kraus operators, with entries in $\{0, 1, 4/5, \pm 3/5\}$, that factorizes exactly through a type II_1 factor. The construction starts from an explicit finite presentation, with 43 generators and 4 417 relators, of a group that maps into the amalgamated double of a Kun–Thom pair. In this presentation Thom’s witness commutator is a nontrivial word of length six, and the recent centralizer-rigidity theorems of Alekseev, Liu and Thom and of Thom imply that every tracial matrix-ultraproduct representation kills it. Rational two-level blocks enforce the relations, and the Choi matrix gives the word a trace the identity cannot have. We also give a support-restricted separation of $1/2$ in half-diamond distance. For a fully referenced version of the channel, any uniform relator tolerance, whose value is still open, converts into explicit global distance bounds and into a linear lower bound on the memory needed for repeated use at small error and bounded exchange. Exact data and independent checks accompany the proofs.

1 The question and the construction

With a freely chosen initial environment, every channel on a finite-dimensional system has an exact finite realization [1]. Requiring the environment to start maximally mixed changes the question: the bath may be as large as we like, but it contributes no purity.

This is a question about Connes embedding [2, 3, 4]. Haagerup and Musat showed that every factorizable channel is a limit of finite matrix-tracial ones exactly when the Connes embedding problem has a positive answer [4, Theorems 3.6 and 3.7]; see also [5, Corollary 3.4] and [6, Corollary 3.11]. Since $\text{MIP}^* = \text{RE}$ [7], factorizable channels outside the closure exist, but the argument does not produce one.

Nonhyperlinear groups have since been found through centralizer rigidity. Building on OpenAI’s nonsofic group [8], Kun and Thom showed that the amalgamated double $G *_H G$ of Kazhdan groups $H < G$, with H infranormal but not normal, is nonsofic, by a centralizer-normalization theorem [9, Theorems A and 4.1], and gave explicit such pairs $\text{EL}_r(\mathbb{F}_q[x_1, \dots, x_d]) < \text{EL}_r(\mathbb{F}_q[x_1^{\pm 1}, \dots, x_d^{\pm 1}]) \rtimes \text{SL}_d(\mathbb{Z})$ for every prime power q and all $r, d \geq 3$ [9, Theorem E]. Thom showed that a positive answer to the centralizer problem of Alekseev and Thom [10, Open Problem 6.2(a)] makes these doubles nonhyperlinear, with an explicit witness commutator [11, Theorems 1.2 and 1.3, Section 5.2]. Liu answered the problem [12, Theorem 1.2]; Alekseev, Liu and Thom proved a stronger internality theorem, the form used here [13, Theorem B], and deduced that the doubles are nonhyperlinear [13, Corollary C]. Section 2 states the one consequence of these recent preprints [11, 13] that we use.

A finite channel can enforce only finitely many relations, so it needs a finitely presented group. These doubles are finitely generated and recursively presented, so Higman’s embedding theorem [14] places them inside finitely presented groups, which are then nonhyperlinear, but the embedding gives no usable presentation. We give an explicit one instead: the amalgamated double of an explicitly presented Kazhdan group $\widehat{G}$ over a finitely generated Kazhdan subgroup $\widehat{H}$, mapping into the double of the Kun–Thom pair with $q = 5$ and $r = d = 3$. Property (T) of $\widehat{G}$ and $\widehat{H}$ can be checked directly with the three-subgroup criterion of Ershov and Jaikin-Zapirain [15], and Thom’s commutator becomes a word of length six. Rational two-level blocks then turn the presentation and the word into channel data, specified entry by entry. To our knowledge this is the first explicit factorizable channel outside the closure of the finite matrix-tracial factorizations.

It differs from earlier explicit examples in both directions. Nonfactorizable channels lie outside the closure but have no tracial realization at all [3]; channels that need an infinite ancilla for an exact factorization can still be approximated by finite ones [16, 17]. The channel here has an exact tracial factorization and no finite matrix-tracial approximation. The appendices give the external input (Appendix A), the complete presentation and its concrete image (Appendices B and C), the word arithmetic and deletion certificates (Appendices D and E), the channels and the global exclusion (Appendices F and G), the quantitative bounds and the memory argument (Appendices H–K), and the data format (Appendix L).

Notation. Write $M_d = M_d(\mathbb{C})$ and $\tau_m = \text{Tr}/m$. Let T_d consist of the channels

$$\Psi(X) = (\text{id} \otimes \tau_m)[U(X \otimes I_m)U^*], \quad U \in U(dm), \quad m \geq 1, \quad (1)$$

and put $K_d = \overline{T_d}$. Finite convex mixtures have the same closure: rational mixture weights are obtained by repeating matrix blocks, and arbitrary weights follow by approximation [6, Proposition 3.8(ii)]. The class Fact_d of factorizable channels, introduced by Anantharaman-Delaroche [18] and characterized in the unitary form used here by Haagerup and Musat [3, Theorem 2.2], allows a finite von Neumann algebra with a faithful normal tracial state in place of M_m . Here “finite” is the operator-algebraic notion; the algebra need not be finite-dimensional. In the notation of Refs. [5, 6], $T_d = \mathcal{FM}_{\text{matr}}(d)$, its convex hull is $\mathcal{FM}_{\text{fin}}(d)$, and $\text{Fact}_d = \mathcal{FM}(d)$. Every channel in K_d is factorizable (Lemma G.1), and by Ref. [4, Theorem 3.6], K_d consists of the factorizable channels with an ancilla that embeds trace-preservingly into the ultrapower of the hyperfinite II_1 factor, the qa-factorizable channels in the terminology of Ref. [19].

We use the normalized Choi state and two channel distances,

$$J(\Phi) = (\Phi \otimes \text{id})(|\Omega_d\rangle\langle\Omega_d|), \quad |\Omega_d\rangle = d^{-1/2} \sum_{k=0}^{d-1} |kk\rangle, \quad (2)$$

$$d_J(\Phi, \Psi) = \frac{1}{2} \|J(\Phi) - J(\Psi)\|_1, \quad d_\diamond(\Phi, \Psi) = \frac{1}{2} \|\Phi - \Psi\|_\diamond. \quad (3)$$

Thus $d_J \leq d_\diamond \leq d d_J$. Their distances to K_d are denoted $\Delta_J(\Phi)$ and $\Delta_\diamond(\Phi)$. All bath 2-norms are normalized: $\|x\|_2^2 = \tau(x^*x)$.

Theorem 1 (Finite presentation and rational channels). *The finite rules in Section 2 define a group $\widehat{D}$ with 43 generators and 4 417 relators, and a named element $j \neq 1$, such that every homomorphism from $\widehat{D}$ into a tracial matrix ultraproduct sends j to the identity. In particular, $\widehat{D}$ is nonhyperlinear.*

The channel construction in Section 3 gives channels Φ_1 , Φ_2 , and Φ_{15} in dimensions 31 981, 31 982, and 32 768, respectively, with

$$\Phi_\nu \in \text{Fact}_{d_\nu} \setminus K_{d_\nu}, \quad \Delta_J(\Phi_\nu) > 0. \quad (4)$$

Each has Choi rank 11 230 and Kraus entries in $\{0, 1, 3/5, 4/5, -3/5\}$. The channel Φ_{15} acts on exactly fifteen system qubits and has 64 748 nonzero Kraus entries.

By Ref. [4, Theorem 3.6], no ancilla for these channels embeds trace-preservingly into the ultrapower of the hyperfinite II_1 factor; Section 4 exhibits an explicit ancilla, the separable II_1 factor $L(\Gamma * F_2)$. Through its first part, the whole theorem rests on the external theorem stated in Section 2; the presentation, the word calculations, the deleted equations and the rational matrices are verified by the accompanying programs. We also keep the original fully referenced channel Φ_{ref} on M_{44086} , whose extra reference positions serve the quantitative estimates of Section 6.

2 A word that matrix approximations must lose

A group is *Kazhdan* if it has property (T). For a subgroup $H < G$, its compression semigroup is

$$P_H = \{t \in G : tHt^{-1} \leq H\}.$$

We use the following consequence of Theorem B of Ref. [13] and Theorem 1.2 of Ref. [11]. Both are recent preprints, unrefereed at the time of writing, and this statement is the only input taken from them:

If H and G are Kazhdan and P_H generates G , then every homomorphism $\pi : G \rightarrow \text{U}(\mathcal{M})$ into a tracial matrix ultraproduct satisfies $\pi(g)(\pi(H)' \cap \mathcal{M})\pi(g)^* = \pi(H)' \cap \mathcal{M}$ for every $g \in G$.

No injectivity or trace-preservation hypothesis is imposed on π .

2.1 The finite rules

Let $C = (0, 1, 2, 3, -1, -2, -3)$. Start with generators f_r , where $f \in \{a, b, c\}$ and $r \in C$. Impose fifth powers, commutation within each family, and

$$[[f_r, g_s], f_u] = [[f_r, g_s], g_u] = 1 \quad ((f, g) = (a, b), (b, c), (c, a); r, s, u \in C), \quad (5)$$

with $[x, y] = xyx^{-1}y^{-1}$. Add six generators t_{ij} , $1 \leq i \neq j \leq 3$, with

$$[t_{ij}, t_{kl}] = 1 \quad (i \neq l, j \neq k), \quad [t_{ij}, t_{jk}] = t_{ik} \quad (i, j, k \text{ distinct}). \quad (6)$$

Trivial and duplicate commuting relations are omitted. For the cyclic successor and predecessor f^+ and f^- of f , set

$$B_f(r, s) = [[f_r, (f^+)_0], [(f^-)_s, f_0]].$$

For $\epsilon = \pm 1$, impose

$$t_{ij}^\epsilon f_r t_{ij}^{-\epsilon} = \begin{cases} B_f(\epsilon i, j), & r = j, \\ B_f(-\epsilon i, -j), & r = -j, \\ f_r, & r = 0 \text{ or } |r| \neq j. \end{cases} \quad (7)$$

Finally put $\zeta_i = (t_{ik} t_{ki}^{-1} t_{ik})^2$, where $k = i + 1$ cyclically, and impose $f_{-i} = \zeta_i f_i \zeta_i^{-1}$. Call the resulting group $\widehat{G}$ and define its *actual subgroup*

$$\widehat{H} = \langle f_r : f \in \{a, b, c\}, r = 0, 1, 2, 3 \rangle < \widehat{G}.$$

These finite rules supply the hypotheses of the external theorem. Any two root families generate a finite class-two group. After removing common invariant vectors, their fixed spaces have angle cosine at most

$1/\sqrt{5} < 1/2$. The three-subgroup criterion of Ref. [15] therefore gives property (T) for the root groups. The six action generators present $\text{St}_3(\mathbb{Z})$, which is Kazhdan [15]. Both signs in Eq. (7) make the full root subgroup normal, and property (T) passes to extensions [20, Proposition 1.7.6], so $\widehat{G}$ is Kazhdan. The subgroup $\widehat{H}$ is a quotient of the positive-root presentation and is Kazhdan as well. Each of the six t_{ij} , taken with exponent +1, compresses $\widehat{H}$ into itself; the naming relations show that they generate $\widehat{G}$ together with $\widehat{H}$. Appendix B gives the fixed-space calculation, the extension argument, and the exact enumeration conventions.

2.2 A concrete image and the surviving word

Let

$$R_+ = \mathbb{F}_5[x_1, x_2, x_3], \quad R = \mathbb{F}_5[x_1^{\pm 1}, x_2^{\pm 1}, x_3^{\pm 1}], \quad G_0 = \text{SL}_3(R) \rtimes \text{SL}_3(\mathbb{Z}).$$

The second factor acts on exponent columns by monomial substitutions. Map a_r, b_r, c_r to elementary matrices in positions $(1, 2), (2, 3), (3, 1)$ with coefficient z_r , where $z_0 = 1$ and $z_{\pm i} = x_i^{\pm 1}$, and map t_{ij} to the action matrix $I + E_{ij}$. All defining relations hold. For $t = t_{12}$ and $h = a_2$, the image of $\widehat{H}$ has polynomial entries, while the image of $t\widehat{H}t^{-1}$ has entries in $\mathbb{F}_5[x_1, x_1x_2, x_3]$. The coefficient x_2 of h is not in that subring. Thus $h \notin t\widehat{H}t^{-1}$. This is the explicit compressor of Kun and Thom [9, proof of Theorem E], also used by Thom [11, Section 5.2].

Form $\widehat{D} = \widehat{G} *_H \widehat{G}$ by identifying the twelve positive-root generators. Write t_0, t_1 for the two copies of t , and adjoin the redundant name

$$j = [h, t_1 t_0^{-1}], \quad (8)$$

the commutator from Thom's proof of his Theorem 1.3 [11, Section 5.2], written in the presented group. This gives 43 generators and 4 417 distinct freely reduced relators, of total length 41 953 and maximum length 19. The signed generator numbers of the witness word are $(3, 37, -22, -3, 22, -37)$; its new generator number is 43.

The two concrete maps extend to

$$D_0 = G_0 *_H G_0, \quad H_0 = \text{SL}_3(R_+) \times \{I\}.$$

There the image of j is

$$(ht)_1 (t^{-1}h^{-1}t)_0 (t^{-1})_1. \quad (9)$$

The outside syllables have nonidentity monomial action. The middle one has coefficient $-x_2/x_1$ in position $(1, 2)$. All three lie outside H_0 , so the reduced-word theorem for an amalgam makes j nontrivial; this is Thom's normal-form argument [11, Section 5.2] carried out in D_0 . Let Γ be the image of $\widehat{D}$ in D_0 .

Now, following Thom's proof [11, Section 5.2], take any matrix-ultraproduct representation π of $\widehat{D}$ and put $A = \pi(\widehat{H})' \cap \mathcal{M}$. The element $c = t_0^{-1}t_1$ centralizes the shared subgroup: both copies implement the same compression on it. Hence $\pi(c) \in A$. The external normalization theorem gives

$$\pi(t_1 t_0^{-1}) = \pi(t_0 c t_0^{-1}) \in A.$$

It commutes with $\pi(h)$, and $\pi(j) = I$. The same assertion holds for representations of Γ by composition with $\widehat{D} \rightarrow \Gamma$. No closure property of hyperlinear groups under arbitrary quotients is needed.

3 From the presentation to rational matrices

The computation uses Γ , not a presumed solution to the word problem of $\widehat{D}$. Laurent-polynomial arithmetic decides multiplication in G_0 and membership in H_0 . Combining adjacent syllables and absorbing H_0 -syllables

decides the word problem in D_0 . To compare two words, reduce their quotient; reduced syllable lists need not be unique.

Collect the identity, the signed generators, and the relator prefixes as labels, identifying equal elements of Γ . A prefix step supplies an equation $xy = z$, represented by a triple (x, y, z) . Include a generator-inverse equation for each positive generator, then remove duplicate triples and equations with an identity input. The original table has $|\mathcal{L}| = 11\,230$ labels and $|\mathcal{T}_0| = 16\,428$ triples. Label 0 is the identity; label 85 is the named element j .

Removing redundant equations. Retain all 43 positive-generator inverse equations. If $y^\vee$ denotes the label of the inverse of a signed generator, then

$$xy = z, \quad yy^\vee = 1 \implies zy^\vee = x.$$

Whenever both multiplication equations occur, one is redundant. This removes 408 triples. Another 30 triples merely reverse a retained inverse equation. The remaining table $\mathcal{T}$ has 15 990 triples. Each deleted equation is supplied as a product of conjugates of *retained* equations; no deletion depends on another deletion. Every original relator is therefore still enforced, and every label still occurs in the retained table.

The blocks. Choose a list B of singleton reference labels containing 0. For each reference position p with label g , set $(A_g)_{pp} = 1$. For the n th retained triple (x, y, z) , use positions $p = |B| + 2n, p + 1$ and put

$$(A_0)_{p,p} = \frac{3}{5}, \quad (A_y)_{p,p+1} = \frac{4}{5}, \quad (A_x)_{p+1,p} = \frac{4}{5}, \quad (A_z)_{p+1,p+1} = -\frac{3}{5}. \quad (10)$$

All other entries vanish. The channel is

$$\Phi_B(X) = \sum_{g \in \mathcal{L}} A_g X A_g^*. \quad (11)$$

Taking $B = (0)$ gives Φ_1 ; taking $B = (0, 85)$ gives Φ_2 . To obtain Φ_{15} , append 786 identity singleton positions to Φ_2 , assigning all of them to label 0. Equivalently, append 786 identity blocks to the factorizing unitary. This prescription also fixes the action on off-diagonal matrix units involving the added positions; it is not merely a specification of two diagonal channel corners.

Channel	Dimension	Triples	Singletons	Nonzero entries
Original Φ_{ref}	44 086	16 428	11 230	76 942
One reference Φ_1	31 981	15 990	1	63 961
Two references Φ_2	31 982	15 990	2	63 962
Fifteen qubits Φ_{15}	32 768	15 990	788	64 748

Table 1: All four channels have Choi rank 11 230 and common Kraus denominator five. In the final row, 787 singleton positions have label 0 and one has label 85.

General factorization and spectrum. Let λ be the regular representation of Γ , with its canonical trace. The block associated with (x, y, z) is

$$\begin{pmatrix} (3/5)I & (4/5)\lambda(y) \\ (4/5)\lambda(x) & -(3/5)\lambda(z) \end{pmatrix} = \text{diag}(I, \lambda(x)) \begin{pmatrix} 3/5 & 4/5 \\ 4/5 & -3/5 \end{pmatrix} \otimes I \text{diag}(I, \lambda(y)). \quad (12)$$

It is unitary because $xy = z$. Its direct sum with the singleton blocks is $U = \sum_g A_g \otimes \lambda(g)$. Distinct group elements are orthogonal in the canonical trace, so U realizes Eq. (11) tracially; this is the factorization criterion of Ref. [3, Theorem 2.2], up to their convention of conjugating by u^* . In particular, the channel is unital and trace preserving.

The Kraus supports are nonempty and pairwise disjoint. Thus the Choi rank is $|\mathcal{L}|$, and its nonzero eigenvalues are

$$\frac{b_g}{25d}, \quad b_g = \sum_{(p,q,n) \in \text{kraus}[g]} n^2. \quad (13)$$

For Φ_{15} the common spectral denominator is 819 200. The package includes the complete weight histograms. These are exact spectral data, not estimates of the global separation.

4 Why one identity reference is enough

The original construction gave every label a singleton. That makes extraction transparent, but it is unnecessary for an exact realization. Once the identity label is unitary, the row and column norms of a two-level block force the other labels to be unitary too.

Lemma 2 (Extraction from Choi support). *Let Φ_B be a channel constructed above, and let $\Pi = \text{supp } J(\Phi_B)$. If a unitary $V \in M_d \otimes N$ in a finite von Neumann algebra with faithful tracial state implements a channel Ψ satisfying $\text{supp } J(\Psi) \leq \Pi$, then*

$$V = \sum_g A_g \otimes v_g$$

for unitaries $v_g \in N$. The normalized labels $w_g = v_0^ v_g$ satisfy $w_0 = I$ and every retained multiplication equation. If $\Psi = \Phi_B$, then also $\tau(v_g v_h^*) = \delta_{g,h}$.*

Proof. The range of Π is the span of the vectorized Kraus matrices. Every vector orthogonal to that span has zero quadratic form in $J(\Psi)$; the corresponding linear combination of the bath blocks of V has zero 2-norm and vanishes. Therefore $V = \sum_g A_g \otimes v_g$.

The identity singleton makes v_0 unitary. Write $a = 3/5$ and $b = 4/5$. Each two-level block of V has the form

$$\begin{pmatrix} av_0 & bv_y \\ bv_x & -av_z \end{pmatrix}.$$

Its first row and first column imply $v_y v_y^* = I$ and $v_x^* v_x = I$. In a finite tracial algebra an isometry or coisometry is unitary. The second column then gives $v_z^* v_z = I$. Every label occurs in a block or a singleton, so all v_g are unitary. Row orthogonality gives

$$v_0 v_x^* = v_y v_z^*, \quad v_z = v_x v_0^* v_y,$$

and hence $w_x w_y = w_z$. Finally, when the Choi states are equal, comparison of coefficients in the orthogonal basis $(|A_g\rangle\rangle)_g$ gives the stated trace identities, as in Ref. [3, Theorem 2.2]. $\square$

Proposition 3 (Presentation-to-channel principle). *Suppose a finite presentation has a named word j killed by every matrix-ultraproduct representation, and a concrete quotient in which $j \neq 1$. Take as labels finitely many pairwise distinct elements of that quotient, including the identity, the images of the generators and of their inverses, and the image of j , and as triples multiplication equations $xy = z$ that hold there. The*

resulting channel lies in $\text{Fact}_d \setminus K_d$ whenever the retained equations imply every presentation relator and generator-inverse equation and identify the label of j with the word j , every label is covered by a block or singleton, and an identity singleton is present.

Proof. General factorization is Eq. (12); it needs the labels to be distinct and the triples to be valid in the quotient. If finite matrix-tracial channels converged to the target, their bath-block ultraproduct would give an exact unitary factorization of the target in a tracial matrix ultraproduct. Compare the implication (2) $\Rightarrow$ (1) of Ref. [4, Theorem 3.6], which is proved there through condition (3), with a factorization through the ultrapower of the hyperfinite factor; here the matrix ultraproduct is taken directly. The system dimension is fixed, so there are only finitely many blocks and Choi entries to pass to the limit.

Lemma 2 gives unitary labels satisfying the presentation. The representation therefore sends j to I . But the identity and j are distinct labels, so

$$1 = \tau(w_j) = \tau(v_0^* v_j) = 0,$$

a contradiction. The approximating unitaries were never assumed to have the prescribed block form; that form is forced only in the exact limit. $\square$

Encoding operator relations in second moments has precedents. Musat and Rørdam force linear relations among unitaries through their trace correlations, following Regev, Slofstra, and Vidick [16, Proposition 3.2], and turn them into a Schur-multiplier channel that needs a type II_1 ancilla. Mori uses correlation matrices of unitaries to force a product relation among symmetries that has no finite-dimensional solution [17]. Passing from a presentation to an object in which a killed word must survive is Slofstra's method for nonlocal games [21]; see also Ref. [19] for games and factorizable maps. Proposition 3 applies these ideas to group relations in a channel.

The deletion certificates verify the hypotheses of Proposition 3 for all three reduced channels. Closedness of K_d gives a positive distance. This completes the proof of Theorem 1.

Replacing the bath $L(\Gamma)$ by $L(\Gamma * F_2)$, where F_2 is the free group on two generators, preserves the trace table and gives a separable II_1 factor (Appendix G). Since $\Phi_v \notin K_{d_v}$, the implication (1) $\Rightarrow$ (2) of Ref. [4, Theorem 3.6] shows that this factor is not Connes-embeddable; compare the existential version in Ref. [6, Corollary 3.11]. At the level of groups this is the standard correspondence between hyperlinear groups and embeddable group von Neumann algebras [22, Proposition 7.1]. Liu's manuscript draws the corresponding non-embedding for the group von Neumann algebra of a wreath product from its Theorem 1.1 [12].

5 An elementary separator on the exact support face

The global obstruction concerns limits of matrix representations. A simpler argument already explains what happens to *exact* finite representations, without property (T) or ultraproduct rigidity: the finite-dimensional form of Thom's observation that a compressor always normalizes a finite image of the subgroup [11, Remark 4.4], combined with his argument for the witness [11, Section 5.2].

Lemma 4 (Finite-dimensional commutants). *If $H < G$, $tHt^{-1} \leq H$, and $D = G *_H G$, then every homomorphism $\pi : D \rightarrow \text{U}(m)$ kills $[h, t_1 t_0^{-1}]$ for every $h \in H$.*

Proof. Put $A = \pi(H)' \cap M_m$. Compression of H gives $\pi(t_0)A\pi(t_0)^* \supseteq A$. Conjugation preserves vector-space dimension, so this inclusion is equality. As before, $t_0^{-1}t_1$ centralizes H ; thus $\pi(t_1 t_0^{-1}) \in A$ and it commutes with $\pi(h)$. $\square$

For a target channel with support projection Π , define

$$T_d^\Pi = \{\Psi \in T_d : \text{supp } J(\Psi) \leq \Pi\}.$$

This is a restricted comparison class. Its closure is not being identified with $K_d \cap \{\text{supp } J \leq \Pi\}$.

Theorem 5 (Support-restricted separators). *For any channel in Table 1, put $s_g = \|A_g\|_{\text{HS}}^2$. Then*

$$\inf_{\Psi \in T_d^\Pi} d_J(\Phi, \Psi) \geq \frac{\sqrt{s_0 s_{85}}}{d}. \quad (14)$$

For Φ_{ref} , Φ_2 , and Φ_{15} , which retain separate identity and witness singletons, one also has

$$\inf_{\Psi \in T_d^\Pi} d_\diamond(\Phi, \Psi) \geq \frac{1}{2}. \quad (15)$$

The last bound is witnessed by one unentangled input and one two-outcome measurement.

Proof. Let $\Psi \in T_d^\Pi$. Lemma 2 gives unitary labels satisfying the retained equations. The deletion certificates (Proposition E.1) and the prefix lemma (Lemma D.1) turn these into a representation of $\widehat{D}$, and Lemma 4 then gives $v_{85} = v_0$. The normalized vectors $|a_g\rangle = |A_g\rangle/\sqrt{s_g}$ are orthogonal. The target Choi state has zero cross entry between $|a_0\rangle$ and $|a_{85}\rangle$; the comparison has cross entry $\sqrt{s_0 s_{85}}/d$, while their two diagonal entries agree. The projection onto $(|a_0\rangle + |a_{85}\rangle)/\sqrt{2}$ proves Eq. (14).

Let p_0, p_j be the two singleton system positions and input $|+\rangle = (|p_0\rangle + |p_j\rangle)/\sqrt{2}$. The target output is the equal incoherent mixture of those positions. Since $v_j = v_0$, every supported comparison leaves $|+\rangle\langle+|$ unchanged. Their output half trace distance is $1/2$; measuring $|+\rangle\langle+|$ attains this difference. Continuity extends both bounds to the closure of the restricted class. $\square$

Rational mixtures of supported finite factorizations are again supported finite factorizations, obtained by repeating blocks, so both bounds hold against rational mixtures and, by approximation, against all finite mixtures, which lie in the closure.

For Φ_{15} , the integer weights are $b_0 = 163\,972$ and $b_{85} = 41$. Equation (14) is therefore strictly larger than $1/316$, as checked by squaring and comparing integers. The corresponding rational lower bounds for Φ_1 and Φ_2 are $1/527$ and $1/329$.

These numerical separators do not evaluate Δ_J or $\Delta_\diamond$. A nearby finite-tracial channel can have small nonzero Choi weight outside Π . Projecting that weight away is not known to preserve a unitary finite-tracial realization. The support hypothesis cannot be removed from Theorem 5 by continuity alone.

6 Quantitative extraction and the remaining tolerance

The universal-kernel result has a uniform finite-error consequence. There is a rational $e_0 \in (0, 1]$ such that, in every matrix dimension and for every unitary assignment to the 43 positive generators,

$$\max_{r \in \mathcal{R}} \|\varphi(r) - I\|_2 \leq e_0 \implies \|\varphi(j) - I\|_2 \leq \frac{1}{2}. \quad (16)$$

Otherwise assignments with vanishing relator defects and witness displacement exceeding $1/2$ would yield a contradicting ultraproduct representation. This is the standard compactness argument; Liu's manuscript draws the same kind of uniform consequence for its group [12, Theorem 1.1]. It proves existence, not a numerical value of e_0 .

Theorem 6 (Global bounds from a relator tolerance). *For the original fully referenced channel Φ_{ref} , every tolerance satisfying Eq. (16) gives*

$$\Delta_J(\Phi_{\text{ref}}) \geq \frac{e_0^2}{65\,188\,028\,416}, \quad (17)$$

$$\Delta_\diamond(\Phi_{\text{ref}}) \geq \frac{e_0^2}{1\,478\,656}. \quad (18)$$

More generally, for a fully referenced channel built by the same block rule, the denominators are $4096\ell_*^2 d$ and $4096\ell_*^2$ when the maximum relator length is $\ell_* \geq 2$ and every relator path and generator inverse is encoded by the triangle table.

The bounds come from the following extraction; Appendix I gives each estimate. Let U implement an arbitrary finite-tracial comparison. A singleton bath block B_g is a contraction. Small probability of leaving its system position gives a unitary polar completion v_g with $\|B_g - v_g\|_2 \leq k$, where $k = \sqrt{d}\delta$ for Choi error δ , or $k = \sqrt{\eta}$ for half-diamond error η . A zero-target Gram constraint then places each triangle entry within $3k$ of its prescribed scalar multiple of v_g . In the half-diamond case this uses a probe supported on two input positions, rather than the uniform d -dimensional Choi probe.

Unitarity controls the row product from outside each two-level block. Row orthogonality therefore gives triangle defects below $32k$, and following the original relator paths gives

$$\max_{r \in \mathcal{R}} \|\varphi(r) - I\|_2 \leq 64\ell_* k.$$

The identity and witness references also give $|\tau(v_0^* v_j)| \leq 2k^2 + 2k$. At $k \leq e_0/(64\ell_*)$, this contradicts Eq. (16). The argument covers arbitrary finite baths, with no support restriction.

The smaller channels do not automatically inherit these constants: they no longer have a separate singleton for every label. We retain Φ_{ref} precisely to keep this quantitative route available.

A Kazhdan angle constant cannot simply be substituted for e_0 . For example, assign one generator $\text{diag}(1, z_m)$ and all others I_2 , where

$$z_m = \frac{m^2 - 1 + 2mi}{m^2 + 1}.$$

The relator defects tend to zero, but the lazy conjugation gap off the coordinate commutant is $1/[43(m^2 + 1)]$. The witness also approaches the identity, so this is not a counterexample to the group theorem. It shows why quantitative lifting or normalization, not the exact group's spectral gap alone, is needed.

7 Purity and memory in causal repeated use

The separation also has a resource interpretation. Use the causal model of Ref. [23, Definition 2.3]: each output is released before the next input arrives; all retained registers count toward a memory of Q qubits; imported states are fixed and independent; and discarded registers never return. Write $P = Q - S(\beta_0)$ for initial purity, C for the sum of imported qubits minus their entropy, and J for the number of replacement slots. An export and its replacement count once. Error ϵ is the largest half trace distance of the complete experiment, over adaptive observers with quantum references.

For any channel in Table 1, the purity converse [23, Theorem 3.2(b)] gives

$$P + C \geq \frac{n(\Delta_J(\Phi) - \epsilon)_+^2}{2 \ln 2}. \quad (19)$$

Appendix K reproduces the local tracialization and entropy ledger needed for this inequality. It imposes no restriction on Q or J .

The fully referenced channel has a pure input-reference probe whose output is flat of rank $r = 11\,230$; it also has maximal complementary entropy $\log_2 r$. Proposition 8.7 of Ref. [24] sharpens the earlier flat-probe converse [23, Theorem 5.8] to $P + C < 3Q + 6$ at error at most $1/16$ and exchange $J \leq n \log_2 r/2$. Combining the two inequalities yields

$$Q \geq \frac{n(\Delta_J(\Phi_{\text{ref}}) - \epsilon)_+^2}{6 \ln 2} - 2. \quad (20)$$

In particular,

$$Q \geq \frac{n\Delta_J(\Phi_{\text{ref}})^2}{24 \ln 2} - 2, \quad \epsilon \leq \min\{1/16, \Delta_J(\Phi_{\text{ref}})/2\}. \quad (21)$$

The exchange restriction is not claimed optimal outside K_d .

The two-reference channels Φ_2 and Φ_{15} retain a flat rank-two probe: the input $|+\rangle$ of Theorem 5. Thus Eq. (20) holds for each of them with its own Δ_J under the more restrictive budget $J \leq n/2$. We do not transfer the original rank-11 230 probe to a channel from which its reference positions have been removed.

8 Data and open questions

The package contains the literal presentation, the original label table, all reduced sparse Kraus matrices, and a free-group certificate for every deleted multiplication equation. The reduced instances are reconstructed independently, by deleting and reindexing slots of the original channel. The checks cover all labels, all retained equations, both Kraus normalization identities, the Choi weights and the integer arithmetic in the restricted bounds, and deliberate corruptions are rejected. They do not formalize the external rigidity theorems or the analytical estimates; Appendix L records which statements are finite checks and which rest on external theorems.

Two questions remain. The first is a number: an evaluated relator tolerance in Eq. (16) would bound $\Delta_J(\Phi_{\text{ref}})$ explicitly, and a quantitative support-leakage theorem would do the same for the reduced channels, whereas $1/316$ and $1/2$ hold only against supported comparisons. The second is size: fifteen system qubits are not claimed to be minimal, and no efficient gate implementation is given. A finite pure environment still realizes the channel exactly, while by Eq. (19) with $n = 1$ any finite environment that realizes it to within ϵ must start with purity at least $(\Delta_J - \epsilon)_+^2/(2 \ln 2)$ bits; the maximally mixed environment is the extreme case.

AI assistance. Generative AI was used in mathematical exploration, code development, checking and writing. The authors are responsible for the content.

A Conventions and external inputs

Products compose from right to left and $[x, y] = xyx^{-1}y^{-1}$. The normalized trace on $M_m(\mathbb{C})$ is $\tau_m = \text{Tr}/m$; $\|x\|_2^2 = \tau_m(x^*x)$, while $\|x\|_{\text{HS}}^2 = \text{Tr}(x^*x)$ denotes the unnormalized Hilbert–Schmidt norm. For a fixed nonprincipal ultrafilter ω on $\mathbb{N}$, a tracial matrix ultraproduct is

$$\mathcal{M} = \prod_{\omega} (M_{m_n}, \tau_{m_n}) = \frac{\{(x_n) : \sup_n \|x_n\|_{\infty} < \infty\}}{\{(x_n) : \lim_{\omega} \|x_n\|_{2, m_n} = 0\}}. \quad (\text{A.1})$$

It has faithful trace $\tau((x_n)_{\omega}) = \lim_{\omega} \tau_{m_n}(x_n)$. A group is hyperlinear if it admits an injective homomorphism into the unitary group of such an ultraproduct. Our universal-kernel statement quantifies over *all* homomorphisms, not just injective or trace-preserving ones.

The property-(T) inputs are the three-subgroup criterion and the Steinberg-group theorem of Ref. [15]. For the matrix-ultraproduct step we use the following external result.

Theorem A.1 (External normalization input: Ref. [11, Theorem 1.2] with Ref. [13, Theorem B]). *Let $H < G$ be Kazhdan groups and suppose that $P_H = \{t \in G : tHt^{-1} \leq H\}$ generates G . For every homomorphism $\pi : G \rightarrow \text{U}(\mathcal{M})$,*

$$\pi(g)(\pi(H)' \cap \mathcal{M})\pi(g)^* = \pi(H)' \cap \mathcal{M} \quad (g \in G). \quad (\text{A.2})$$

Theorem B of Ref. [13] supplies internality of the relevant relative commutants; Theorem 1.2 of Ref. [11] then gives Eq. (A.2). Property (T) supplies the conjugation spectral gap needed for internality. Neither finite testing nor same-dimension repair of approximate representations is substituted for them.

The centralizer problem answered in Ref. [13] is Open Problem 6.2(a) of Ref. [10]; the proofs in Ref. [13] are based on Liu’s manuscript, whose Theorem 1.2 answers it [12]. Liu’s manuscript is not on arXiv and is cited from an archived copy for provenance; the input used here is Theorem B of Ref. [13]. Neither Theorem B of Ref. [13] nor Theorem 1.2 of Ref. [11] assumes that the dimensions m_n grow. If they are bounded along ω , the ultraproduct is a matrix algebra, and the conclusion used below, $\pi(j) = I$, also follows from Lemma H.1.

B The literal finite presentation

B.1 Three root families

For an ordered finite set C of size m , let $\Delta(C)$ have generators f_r with $f \in \{a, b, c\}$ and $r \in C$, subject to

$$f_r^5 = 1, \quad (\text{B.1})$$

$$[f_r, f_s] = 1 \quad (r < s \text{ in the chosen order}), \quad (\text{B.2})$$

$$[[f_r, g_s], f_u] = [[f_r, g_s], g_u] = 1 \quad ((f, g) = (a, b), (b, c), (c, a); r, s, u \in C). \quad (\text{B.3})$$

The order in Eq. (B.2) fixes enumeration only.

Lemma B.1. *Every $\Delta(C)$ has property (T).*

Proof. A single family generates an elementary abelian group of exponent five. For consecutive families $X = \langle f_r \rangle$ and $Y = \langle g_s \rangle$, the mixed commutators are central in $\langle X, Y \rangle$ by Eq. (B.3), and $[f_r, g_s]^5 = [f_r^5, g_s] = 1$. Collection expresses every element using at most $2m + m^2$ generators or central commutators, with exponents

from 0 to 4. Thus the universal two-family group has order at most 5^{2m+m^2} . Additional relations elsewhere can only produce a quotient of this finite group.

Remove the invariant part of a unitary representation of the two-family group and decompose it into irreducibles. In an irreducible summand, central commutators act by scalars, defining a bicharacter

$$\chi : X \times Y \longrightarrow \{z : z^5 = 1\}.$$

If χ is trivial, the summand is a nontrivial character of an abelian quotient, and at least one of its X - or Y -fixed spaces is zero. Otherwise

$$Y_0 = \{y \in Y : \chi(x, y) = 1 \text{ for every } x \in X\}$$

has index at least five. If v is X -fixed and $y \notin Y_0$, conjugation by a suitable x multiplies $\langle v, yv \rangle$ by a nontrivial fifth root of unity while leaving it unchanged. Hence this coefficient vanishes. Averaging over Y gives

$$\|P_Y v\|^2 = \langle v, P_Y v \rangle \leq \frac{|Y_0|}{|Y|} \|v\|^2 \leq \frac{1}{5} \|v\|^2.$$

The same bound holds in a general representation without nonzero $\langle X, Y \rangle$ -invariant vectors. The projections P_X and P_Y commute with the projections onto the irreducible summands, so an X -fixed vector is an orthogonal sum $v = \sum_i v_i$ of X -fixed vectors in the summands, and $\|P_Y v\|^2 = \sum_i \|P_Y v_i\|^2 \leq \frac{1}{5} \|v\|^2$. This is the supremum over summands used in the proof of Ref. [15, Corollary 4.7], and removing the common invariant part restricts to the representations considered there. The cosine between the two fixed spaces, after removing their common invariant part, is therefore at most $1/\sqrt{5}$. Lifting representations preserves the bound in every quotient. Relative property (T) with respect to each finite family is automatic. The three-subgroup criterion [15], with $1/\sqrt{5} < 1/2$, proves the claim. $\square$

B.2 Action generators and conjugation rules

Let T have six generators t_{ij} for $1 \leq i \neq j \leq 3$, with relations

$$[t_{ij}, t_{kl}] = 1 \quad (i \neq l, j \neq k), \quad (\text{B.4})$$

$$[t_{ij}, t_{jk}] = t_{ik} \quad (i, j, k \text{ distinct}). \quad (\text{B.5})$$

After omitting same-generator commutators and duplicate unordered commuting pairs, there are six commuting relations and six product relations.

Lemma B.2. *These relations present $T \cong \text{St}_3(\mathbb{Z})$, and T is Kazhdan.*

Proof. Put $E_{ij}(n) = t_{ij}^n$. Additivity is immediate. Equation (B.4) gives the commuting-root relations at all integer powers. For distinct i, j, k , the element t_{ik} commutes with both t_{ij} and t_{jk} , so

$$[t_{ij}^u, t_{jk}^v] = t_{ik}^{uv} \quad (u, v \in \mathbb{Z}).$$

These are the remaining Steinberg relations. The maps between T and $\text{St}_3(\mathbb{Z})$ are inverse on generators. Property (T) follows from Ref. [15]. No identification of $\text{St}_3(\mathbb{Z})$ with $\text{SL}_3(\mathbb{Z})$ is used. $\square$

Fix $C = (0, 1, 2, 3, -1, -2, -3)$ and start with $\Delta(C)$ and T . For cyclic successor and predecessor families f^+, f^- , define the word

$$B_f(r, s) = [[f_r, (f^+)_0], [(f^-)_s, f_0]]. \quad (\text{B.6})$$

For every $i \neq j$, $\epsilon \in \{1, -1\}$, $f \in \{a, b, c\}$, and $r \in C$, impose

$$t_{ij}^\epsilon f_r t_{ij}^{-\epsilon} = \begin{cases} B_f(\epsilon i, j), & r = j, \\ B_f(-\epsilon i, -j), & r = -j, \\ f_r, & r = 0 \text{ or } |r| \neq j. \end{cases} \quad (\text{B.7})$$

The signed indices in B_f are coefficient labels, not generator powers. Let $k = i + 1$ cyclically modulo three, set $\zeta_i = (t_{ik} t_{ki}^{-1} t_{ik})^2$, and add

$$f_{-i} = \zeta_i f_i \zeta_i^{-1} \quad (f \in \{a, b, c\}, i = 1, 2, 3). \quad (\text{B.8})$$

Call the resulting group $\widehat{G}$ and define

$$\widehat{H} = \langle a_r, b_r, c_r : r = 0, 1, 2, 3 \rangle < \widehat{G}. \quad (\text{B.9})$$

Proposition B.3. *The groups $\widehat{H}$ and $\widehat{G}$ are Kazhdan. The six t_{ij} compress $\widehat{H}$ into itself and generate $\widehat{G}$ together with $\widehat{H}$.*

Proof. The subgroup $\widehat{H}$ is a quotient of $\Delta(0, 1, 2, 3)$, so it is Kazhdan by Lemma B.1. The subgroup N generated by all 21 root symbols is a Kazhdan quotient of $\Delta(C)$. Both signs in Eq. (B.7) make N normal in $\widehat{G}$. The quotient $\widehat{G}/N$ is a quotient of T , so it has property (T) by Lemma B.2. Property (T) passes to extensions [20, Proposition 1.7.6], so $\widehat{G}$ has it.

For clarity, here is that argument for ordinary unitary representations. Property (T) of the normal subgroup forces almost invariant unit vectors to approach its invariant subspace. This subspace is invariant under $\widehat{G}$, and the quotient has property (T), so it contains a nonzero $\widehat{G}$ -invariant vector.

For $\epsilon = 1$ and $r \geq 0$, the right side of Eq. (B.7) uses only positive labels. Thus $t_{ij} \widehat{H} t_{ij}^{-1} \leq \widehat{H}$. Equation (B.8) recovers every negative root symbol from the positive ones and the action generators. These elements generate $\widehat{G}$. Since $\widehat{H} \subseteq P_{\widehat{H}}$, the whole compression semigroup generates $\widehat{G}$ as well. $\square$

C The computable image and the universal-kernel word

Let

$$R_+ = \mathbb{F}_5[x_1, x_2, x_3], \quad R = \mathbb{F}_5[x_1^{\pm 1}, x_2^{\pm 1}, x_3^{\pm 1}]. \quad (\text{C.1})$$

For $A \in \text{SL}_3(\mathbb{Z})$ define $\sigma_A(x^\nu) = x^{A\nu}$, with exponents written as columns. The semidirect product law is

$$G_0 = \text{SL}_3(R) \rtimes_{\sigma} \text{SL}_3(\mathbb{Z}), \quad (M, A)(N, B) = (M\sigma_A(N), AB). \quad (\text{C.2})$$

With $e_{ij}(z) = I + zE_{ij}$, set

$$a_r \mapsto (e_{12}(z_r), I), \quad b_r \mapsto (e_{23}(z_r), I), \quad c_r \mapsto (e_{31}(z_r), I), \quad (\text{C.3})$$

$$t_{ij} \mapsto (I, I + E_{ij}), \quad (\text{C.4})$$

where $z_0 = 1$ and $z_{\pm i} = x_i^{\pm 1}$.

Lemma C.1. *Equations (C.3)–(C.4) define a homomorphism $\widehat{G} \rightarrow G_0$. For $t = t_{12}$ and $h = a_2$, one has $h \in \widehat{H} \setminus t\widehat{H}t^{-1}$.*

Proof. Fifth powers vanish in characteristic five, and consecutive elementary-root families have central mixed commutators. The integer action matrices satisfy the Steinberg relations. For example, the image of $B_a(r, s)$ is

$$[e_{13}(z_r), e_{32}(z_s)] = e_{12}(z_r z_s);$$

the other families follow by cycling matrix indices. The action $I + \epsilon E_{ij}$ sends x_j to $x_i^\epsilon x_j$ and x_j^{-1} to $x_i^{-\epsilon} x_j^{-1}$, which proves Eq. (B.7). The image of ζ_i is minus identity on coordinates i, k and identity on the remaining coordinate; it inverts x_i , proving Eq. (B.8).

The image of $\widehat{H}$ has polynomial entries. Its conjugate by t_{12} has entries in $\mathbb{F}_5[x_1, x_1 x_2, x_3]$. Every monomial in this subring has first exponent at least its second. The coefficient x_2 in the image of $h = a_2$ has exponent $(0, 1, 0)$ and is not in that subring. This is the compressor in the proof of Theorem E of Ref. [9], also used in Ref. [11, Section 5.2]. Faithfulness of the homomorphism is not required. $\square$

C.1 The double and its enumeration

Form

$$\widehat{D} = \widehat{G} *_H \widehat{G}$$

by taking two copies of the presentation and identifying the twelve positive-root generators. The subgroup $\widehat{H}$ need not have a separate finite presentation: the amalgamated-product universal property only requires its finite generating set for these identifications. Adjoin a named generator j with relator

$$j^{-1}[h, t_1 t_0^{-1}] = 1. \quad (\text{C.5})$$

This is a Tietze transformation and does not change the group. The commutator $[h, t_1 t_0^{-1}]$ and the normal-form argument below are those of Thom's proof of his Theorem 1.3 [11, Section 5.2]; here they are carried out in the presented group and in its concrete image.

The concrete maps agree on $\widehat{H}$, giving a map into

$$D_0 = G_0 *_H G_0, \quad H_0 = \text{SL}_3(R_+) \times \{I\}. \quad (\text{C.6})$$

The image of j has the reduced form

$$(ht)_1 (t^{-1} h^{-1} t)_0 (t^{-1})_1. \quad (\text{C.7})$$

The first and last syllables have nonidentity action matrices. The middle one has matrix $I - (x_2/x_1)E_{12}$ and lies outside H_0 . Thus all syllables are outside the amalgamating subgroup and their sides alternate. The word is nontrivial. In particular $j \neq 1$ in $\widehat{D}$ and in $\Gamma = \text{im}(\widehat{D} \rightarrow D_0)$.

Indices	Generators in enumeration order
1–12	$a_0, a_1, a_2, a_3, b_0, b_1, b_2, b_3, c_0, c_1, c_2, c_3$
13–21	$a_{-1}, a_{-2}, a_{-3}, b_{-1}, b_{-2}, b_{-3}, c_{-1}, c_{-2}, c_{-3}$, copy 0
22–27	$t_{12}, t_{13}, t_{21}, t_{23}, t_{31}, t_{32}$, copy 0
28–36	The nine negative-root symbols in the same order, copy 1
37–42	The six action generators in the same order, copy 1
43	j

Table C.1: A negative integer denotes the inverse of a generator, not a negative coefficient label. For example, generator 14 is a_{-2} in copy 0, whereas -3 is the inverse of a_2 .

The witness word is $(3, 37, -22, -3, 22, -37)$ and its defining relator is $(-43, 3, 37, -22, -3, 22, -37)$. Before deduplication, each copy contributes

$$(21 + 63 + 2058) + 12 + 252 + 9 = 2415$$

relators. The copies share $12 + 18 + 384 = 414$ positive-root relations. Including the new name gives $2 \cdot 2415 - 414 + 1 = 4417$ distinct freely reduced relators.

Family	Number retained
Fifth powers	30
Within-family commutators	108
Class-two relations	3732
Steinberg commuting relations	12
Steinberg product relations	12
Forward and inverse conjugation rules	504
Negative-label names	18
Distinguished-word name	1
Total	4417

Table C.2: The total freely reduced length is 41 953 and the maximum is 19. Free reduction removes adjacent inverse letters only; cyclic conjugates are not identified.

C.2 Universal kernel and a fixed-error tolerance

Proposition C.2. *Every homomorphism $\pi : \widehat{D} \rightarrow \mathcal{U}(\mathcal{M})$ sends j to I . The same is true for the nontrivial image of j in Γ .*

Proof. We follow Thom's proof of his Theorem 1.3 [11, Section 5.2]. Put $A = \pi(\widehat{H})' \cap \mathcal{M}$. Both copies conjugate every $h' \in \widehat{H}$ by t to the same element of the shared subgroup. This holds for the twelve generators by Eq. (B.7), and hence for all their words. Therefore $c = t_0^{-1}t_1$ centralizes $\widehat{H}$, so $\pi(c) \in A$.

Proposition B.3 verifies the hypotheses of Theorem A.1 in copy 0. Consequently

$$\pi(t_1 t_0^{-1}) = \pi(t_0) \pi(c) \pi(t_0)^* \in A.$$

It commutes with $\pi(h)$, giving $\pi(j) = I$. For a representation of Γ , compose with $\widehat{D} \rightarrow \Gamma$. $\square$

Corollary C.3. *There is a rational $e_0 \in (0, 1]$ such that every finite-dimensional unitary assignment φ to the 43 positive generators satisfies*

$$\max_{r \in \mathcal{R}} \|\varphi(r) - I\|_2 \leq e_0 \implies \|\varphi(j) - I\|_2 \leq \frac{1}{2}. \quad (\text{C.8})$$

Proof. If no positive tolerance worked, for each n there would be an assignment in some dimension m_n with relator defects at most $1/n$ and witness displacement greater than $1/2$. The ultraproduct classes satisfy the presentation exactly and give a representation with $\|\pi(j) - I\|_2 \geq 1/2$, contradicting Proposition C.2. Once a positive tolerance exists, choose a smaller positive rational one, at most one. This is the standard compactness argument; compare the uniform statement in Ref. [12, Theorem 1.1]. $\square$

The presentation and the witness threshold are fixed before the dimension is chosen. No dimension cutoff occurs in Eq. (C.8).

D Exact word arithmetic and the original label table

A Laurent polynomial is a finite list of exponent triples in $\mathbb{Z}^3$ with coefficients in $\{1, 2, 3, 4\}$. Combining like terms modulo five and removing zero coefficients gives an exact representation. Integer action matrices remain over $\mathbb{Z}$; they are not reduced modulo five. Besides Eq. (C.2), the inverse rule is

$$(M, A)^{-1} = (\sigma_{A^{-1}}(M^{-1}), A^{-1}). \quad (\text{D.1})$$

A determinant-one matrix is inverted by its adjugate, so no general ring inversion procedure is hidden in this rule.

Membership in H_0 is exactly the condition that the action matrix is I and every nonzero monomial in every matrix entry has nonnegative exponent coordinates. For sufficiency, a polynomial determinant-one matrix has a polynomial adjugate, hence belongs to $\text{SL}_3(R_+)$. Using the full special linear group for H_0 avoids any need to identify it with an elementary subgroup.

An amalgam word is a list of pairs consisting of a side and an element of G_0 . Combine adjacent syllables on the same side, remove identity syllables, and absorb a syllable of H_0 into a neighbor. Each move shortens the list. A final alternating list of length at least two with no syllable in H_0 is nontrivial; a one-syllable list is evaluated in G_0 . Equality of words v, w is tested by reducing $v^{-1}w$. Literal equality of two reduced lists is not used as a general equality criterion.

D.1 Enumeration of labels and triangles

Label 0 is the identity. Visit generators in the numbered order above, each immediately followed by its inverse. Then visit relators in their stored order, and their prefixes from left to right. Assign a new integer label at its first occurrence; reuse an earlier label exactly when the quotient-word test identifies the two elements. The named generator j receives label 85. This label number is distinct from generator number 43.

For a relator $s_1 \cdots s_\ell$, include

$$(\ell(s_1 \cdots s_{k-1}), \ell(s_k), \ell(s_1 \cdots s_k)) \quad (2 \leq k \leq \ell), \quad (\text{D.2})$$

where $\ell(w)$ denotes a label, not word length. Also include $(\ell(s), \ell(s^{-1}), 0)$ for every positive generator. Keep the first occurrence of each ordered triple; identity-input triples may be omitted, though none occur in this instance. There are 37 579 occurrences before deduplication, 16 428 retained triples in $\mathcal{T}_0$, and 11 230 distinct labels.

Lemma D.1. *Unitary labels w_g with $w_0 = I$ satisfying every equation in $\mathcal{T}_0$ define a homomorphism of $\widehat{D}$ by $s \mapsto w_{\ell(s)}$.*

Proof. The generator-inverse equations give $w_{\ell(s^{-1})} = w_{\ell(s)}^{-1}$. Induction along Eq. (D.2) identifies a prefix label with the product of its assigned generator matrices. The final prefix has identity label, so every relator holds. A removed duplicate was already enforced; an identity-input equation is automatic after $w_0 = I$. $\square$

E Certified removal of 438 triangles

The labels and the group presentation do not change in the new reduction. Only redundant multiplication equations are removed. Protect the set

$$\mathcal{F} = \{(\ell(s), \ell(s^{-1}), 0) : s = 1, \dots, 43\}.$$

For a label y of a signed generator, write $y^\vee$ for its inverse label. The original table contains the protected equation in one of the orientations $(y, y^\vee, 0)$ or $(y^\vee, y, 0)$.

For each pair of triples (x, y, z) and $(z, y^\vee, x)$ in $\mathcal{T}_0$, retain the lexicographically smaller triple and delete the larger. The program checks that no protected triple occurs in these pairs. There are 408 such deletions. Next remove each zero-output triple $(x, y, 0)$ outside $\mathcal{F}$ whose reversed triple $(y, x, 0)$ belongs to $\mathcal{F}$; there are 30. All retained triples keep their original order.

E.1 Free-group certificates

Let X_g be a free generator for label $g \neq 0$ and put $X_0 = 1$. A triple $t = (x, y, z)$ has relation word

$$r_t = X_x X_y X_z^{-1}.$$

For a retained relation $xy = z$ and inverse equation $q = X_y X_{y^\vee}$,

$$r_t^{-1} X_x q X_x^{-1} = X_z X_{y^\vee} X_x^{-1}. \quad (\text{E.1})$$

This is an identity in the free group. If the protected inverse equation instead reads $q' = X_{y^\vee} X_y$, use $q = X_y q' X_y^{-1}$. Thus the first family of deletions has certificates with two conjugates of retained equations. A reversed inverse is certified by

$$X_x (X_y X_x) X_x^{-1} = X_x X_y. \quad (\text{E.2})$$

The second family has one cell per certificate.

The stored certificate convention uses integer $g + 1$ for X_g , with label 0 represented by the empty word. Each cell has a retained original triangle index, a sign ± 1 , and a conjugating word. The verifier multiplies all cells, freely reduces the result, and compares it with the relation word of the deleted triple. It also checks that no referenced equation has itself been deleted.

Proposition E.1. *The 15 990 retained equations imply all 16 428 original equations. Every label occurs in the retained table. Consequently Lemma D.1 still holds when only the retained equations are assumed.*

Proof. The 438 certificate identities prove the missing equations. The certificate list and retained indices form a disjoint partition of $\mathcal{T}_0$, and every cell uses a retained equation. Coverage is a finite check of the retained label indices. Applying Lemma D.1 proves the final assertion. $\square$

Coverage, rather than a claimed minimality property, is what the channel proof needs. No assertion is made that this is the smallest sufficient triangle table.

F The channel family and exact extraction

Let $a = 3/5$ and $b = 4/5$. A reference list B contains at least the identity label 0. Put one entry $(A_g)_{pp} = 1$ at each reference position p labeled by g . A triangle (x, y, z) uses two new system positions $p, p + 1$ with

$$(A_0)_{p,p} = a, \quad (A_y)_{p,p+1} = b, \quad (A_x)_{p+1,p} = b, \quad (A_z)_{p+1,p+1} = -a. \quad (\text{F.1})$$

Different blocks have disjoint system positions; all unspecified entries are zero. Set $\Phi(X) = \sum_g A_g X A_g^*$.

The original channel Φ_{ref} uses $\mathcal{T}_0$ and one singleton for every label. The reduced channel Φ_1 uses $\mathcal{T}$ with $B = (0)$; Φ_2 uses $B = (0, 85)$. Their natural dimensions are $1 + 2 \cdot 15990 = 31981$ and $2 + 2 \cdot 15990 = 31982$.

For Φ_{15} append 786 singleton positions with label 0 to Φ_2 . These occupy positions 31982, $\dots$, 32767. The defining Kraus matrix A_0 receives entry one on each added diagonal position. All other Kraus matrices vanish there.

Proposition F.1. *Each channel is unital, trace preserving, and generally tracially factorizable, and has Choi rank 11 230. Its nonzero Choi eigenvalues are $b_g/(25d)$, with b_g the sum of squared integer Kraus numerators for label g .*

Proof. Let λ be the regular representation of Γ . The triangle block

$$\begin{pmatrix} aI & b\lambda(y) \\ b\lambda(x) & -a\lambda(z) \end{pmatrix} = \text{diag}(I, \lambda(x)) \left(\begin{pmatrix} a & b \\ b & -a \end{pmatrix} \otimes I \right) \text{diag}(I, \lambda(y))$$

is unitary, since $xy = z$ and $a^2 + b^2 = 1$. Its direct sum with the reference unitaries is $U = \sum_g A_g \otimes \lambda(g)$. The canonical trace satisfies $\tau(\lambda(g)\lambda(h)^*) = \delta_{g,h}$, giving the stated Kraus channel and a general tracial factorization by the criterion of Ref. [3, Theorem 2.2], up to their convention of conjugating by u^* .

Every Kraus matrix has nonempty support, by coverage, and different labels have disjoint supports. Hence the A_g are linearly independent and Hilbert–Schmidt orthogonal. Therefore

$$J(\Phi) = d^{-1} \sum_g |A_g\rangle\rangle\langle\langle A_g|, \quad \text{spec}_{>0} J(\Phi) = \left\{ \frac{b_g}{25d} : g \in \mathcal{L} \right\}.$$

Each A_g is a weighted partial permutation. In a triangle with nonidentity inputs, $z = x$ would force $y = 1$ and $z = y$ would force $x = 1$, so neither occurs. The possible coincidences $x = y$ and $z = 0$ occupy different rows and columns within their Kraus matrix. Different blocks are disjoint. Thus both normalization identities can also be checked from integer row and column squared sums, each equal to 25. $\square$

F.1 Support determines the bath block array

Let $\Pi = \text{supp } J(\Phi)$. For a unitary realization V in $M_d \otimes N$, write V_{pq} for its bath blocks. The Choi entry at $(p, q), (r, s)$ is $d^{-1} \tau(V_{pq} V_{rs}^*)$.

Lemma F.2. *If $\text{supp } J(\Psi) \leq \Pi$ for the channel implemented by V , then $V = \sum_g A_g \otimes v_g$ for unitaries $v_g \in N$. The labels $w_g = v_0^* v_g$ satisfy $w_0 = I$ and $w_x w_y = w_z$ for every retained triple. If $\Psi = \Phi$, then $\tau(v_g v_h^*) = \delta_{g,h}$.*

Proof. The range of Π is the span of the vectorized A_g . A scalar vector perpendicular to that span has zero quadratic form in $J(\Psi)$, so its corresponding linear combination of bath blocks has zero 2-norm. Faithfulness makes that combination zero. A basis of these linear constraints puts the full block array in the span of the A_g , giving $V = \sum_g A_g \otimes v_g$.

The singleton for label 0 has no other nonzero entry in its row or column, hence v_0 is unitary. Every triangle block is

$$\begin{pmatrix} av_0 & bv_y \\ bv_x & -av_z \end{pmatrix}.$$

Its first row gives $a^2 I + b^2 v_y v_y^* = I$ and its first column gives $a^2 I + b^2 v_x^* v_x = I$. A finite tracial algebra has no proper isometries: if $v^* v = I$, then $I - vv^*$ is a projection of trace zero. Thus v_x, v_y are unitary, and the second column gives $v_z^* v_z = I$. All labels are covered. Row orthogonality gives $v_0 v_x^* = v_y v_z^*$, equivalently $v_z = v_x v_0^* v_y$, and normalization gives the multiplication equation for w .

When the Choi states are equal, comparison in the orthogonal Kraus-vector basis gives $\tau(v_g v_h^*) = \delta_{g,h}$. $\square$

G Excluding every convergent finite-bath sequence

Lemma G.1. *If a channel Φ lies in K_d , it has an exact unitary realization in $M_d \otimes \mathcal{M}$ for a tracial matrix ultraproduct $\mathcal{M}$.*

Proof. Choose finite matrix-tracial factorizations converging to Φ . The system dimension d is fixed. Every bath block is an operator-norm contraction, so its ultraproduct class exists. The finitely many classes form a unitary because the original unitary identities are exact. The trace of each bath-block product converges to its specified Choi entry, giving exactly Φ . Compare the implication (2) $\Rightarrow$ (1) of Ref. [4, Theorem 3.6], which is proved there through condition (3), with a factorization through the ultrapower of the hyperfinite factor; here the matrix ultraproduct is taken directly.

For completeness, finite mixtures give no larger closure; see also Ref. [6, Proposition 3.8(ii)]. For rational mixture weights n_i/N and bath dimensions m_i , choose a common multiple L of the m_i and use $n_i L/m_i$ copies of the i th factorizing unitary. The total bath dimension is NL and its normalized trace gives weight n_i/N . Approximation of real weights completes the reduction. $\square$

Theorem G.2. *Each of Φ_{ref} , Φ_1 , Φ_2 , and Φ_{15} lies in $\text{Fact}_d \setminus K_d$ and has $\Delta_J(\Phi) > 0$.*

Proof. Proposition F.1 gives general factorization. Membership in K_d would give an exact ultraproduct factorization by Lemma G.1. Extract the unitary labels using Lemma F.2. The pruning certificates and the original prefix argument give a representation of $\widehat{D}$. Proposition C.2 forces $w_{85} = I$. But labels 0 and 85 are distinct and the target trace table gives $\tau(w_{85}) = \tau(v_0^* v_{85}) = 0$, contradicting $\tau(I) = 1$. Closedness of K_d in the finite-dimensional space of channels gives positive distance. $\square$

This argument does not assume that the finite approximating unitaries already have the triangle block form. The exact target forces that form after passage to the ultraproduct. In particular, excluding exact finite-dimensional representations alone would not prove the theorem.

G.1 A factor realization

Let F_2 be the free group on two generators. The inclusion $\Gamma < \Gamma * F_2$ preserves the canonical trace, so the same controlled unitary realizes every channel in $L(\Gamma * F_2)$. The free product is countable and has infinite conjugacy classes: suitable reduced words in the two free generators give infinitely many distinct conjugates of any nonidentity word. Thus its group von Neumann algebra is a type II_1 factor with separable predual.

If it embedded trace-preservingly into the hyperfinite ultrapower R^ω , its finite unitary tuples would have simultaneous matrix approximations to arbitrary finite lists of moments. Coordinatewise approximation in finite-dimensional subalgebras of the hyperfinite factor, followed by a diagonal choice, would give a matrix-ultraproduct representation of $\widehat{D}$ retaining the trace $\tau(j) = 0$. This contradicts Proposition C.2. At the level of channels this is the implication (1) $\Rightarrow$ (2) of Ref. [4, Theorem 3.6], and at the level of groups the standard correspondence between hyperlinear groups and embeddable group von Neumann algebras [22, Proposition 7.1]. Liu's manuscript draws the corresponding non-embedding for the group von Neumann algebra of a wreath product from its Theorem 1.1 [12]. By the same implication, no finite von Neumann algebra that serves as an ancilla for one of these channels embeds trace-preservingly into R^ω ; compare Ref. [6, Corollary 3.11]. The nonembeddable algebra is the bath factor, not the finite system algebra M_d .

H The exact-support numerical bounds

The following is the finite-dimensional form of Thom's observation that a compressor normalizes every finite image of the subgroup [11, Remark 4.4], combined with his argument for the witness [11, Section 5.2].

Lemma H.1 (Exact finite-dimensional rigidity). *Let $H < G$ and $tHt^{-1} \leq H$. In $D = G *_H G$, put $j = [h, t_1 t_0^{-1}]$ for $h \in H$. Every homomorphism $\pi : D \rightarrow \mathcal{U}(m)$ sends j to I .*

Proof. For $A = \pi(H)' \cap M_m$, compression gives $\pi(t_0)A\pi(t_0)^* \supseteq A$. The two algebras have the same finite vector-space dimension, so the inclusion is equality. Since $t_0^{-1}t_1$ centralizes H , its image lies in A ; conjugating by $\pi(t_0)$ puts $\pi(t_1 t_0^{-1})$ in A . It commutes with $\pi(h)$. $\square$

The same proof works in a finite direct sum of matrix algebras. For our presentation, the compression assumption follows directly from the twelve positive-root conjugation rules; property (T) is not needed. The dimension argument does not extend to a relative commutant inside a matrix ultraproduct, which can be infinite-dimensional.

Let $T_d^\Pi = \{\Psi \in T_d : \text{supp } J(\Psi) \leq \Pi\}$, where $\Pi = \text{supp } J(\Phi)$. This class is nonempty: assigning every label the identity produces a unitary direct sum of scalar blocks whose Choi support is contained in Π .

Theorem H.2. *Put $s_0 = \|A_0\|_{\text{HS}}^2$ and $s_j = \|A_{85}\|_{\text{HS}}^2$. Then*

$$\inf_{\Psi \in T_d^\Pi} d_J(\Phi, \Psi) \geq \frac{\sqrt{s_0 s_j}}{d}. \quad (\text{H.1})$$

If the two singleton references 0 and 85 are present, then also

$$\inf_{\Psi \in T_d^\Pi} d_\diamond(\Phi, \Psi) \geq \frac{1}{2}. \quad (\text{H.2})$$

Proof. Lemma F.2 gives unitary labels for each $\Psi \in T_d^\Pi$. The original or derived multiplication equations give the finite presentation, and Lemma H.1 gives $v_{85} = v_0$. Let $|a_0\rangle = |A_0\rangle\rangle/\sqrt{s_0}$ and $|a_j\rangle = |A_{85}\rangle\rangle/\sqrt{s_j}$. These vectors are orthogonal. Both Choi states have diagonal entries $s_0/d, s_j/d$ on their span. The target cross entry is zero and the comparison cross entry is $\sqrt{s_0 s_j}/d$. The rank-one effect

$$E = \frac{1}{2}(|a_0\rangle + |a_j\rangle)(\langle a_0| + \langle a_j|)$$

has expectation difference $\sqrt{s_0 s_j}/d$, proving Eq. (H.1).

For the half-diamond assertion, take the two singleton positions p_0, p_j and set $|+\rangle = (|p_0\rangle + |p_j\rangle)/\sqrt{2}$. The target output is $(|p_0\rangle\langle p_0| + |p_j\rangle\langle p_j|)/2$. Since $v_{85} = v_0$, the comparison acts as the identity on this two-dimensional corner and outputs $|+\rangle\langle +|$. The difference has eigenvalues $1/2, -1/2$, so its half trace norm is $1/2$. The projection $|+\rangle\langle +|$ attains this expectation difference. No input-reference entanglement is required. Continuity extends the two bounds to the stated closure. $\square$

Indeed $s_g = b_g/25$, so a positive rational q is certified by $q^2 < b_0 b_{85}/(25d)^2$. For the fifteen-qubit instance the exact comparison is

$$316^2(163972 \cdot 41) > (25 \cdot 32768)^2.$$

The complete comparison uses all finite bath dimensions but an *exact* support restriction. Arbitrarily close Choi states need not have exactly the same support. Projecting onto Π need not preserve a unitary finite-tracial realization. In particular, neither $1/316$ nor $1/2$ is a global separation certificate.

Instance	d	b_0	b_{85}	Strict rational bound for d_J
Original Φ_{ref}	44 086	148 534	41	1/447
Reference-only reduction, two	32 858	148 534	41	3/1000
Reference-only reduction, one	32 857	148 534	16	1/533
Triangle-reduced Φ_1	31 981	144 322	16	1/527
Triangle-reduced Φ_2	31 982	144 322	41	1/329
Fifteen-qubit Φ_{15}	32 768	163 972	41	1/316

Table H.1: All bounds compare with $\overline{T_d^\Pi}$, not with K_d . The two reference-only variants are kept in the baseline part of the package. Each displayed bound is verified by integer arithmetic.

I Global quantitative extraction in two metrics

Throughout this section the channel is fully referenced: each label has its own singleton position p_g . Every relator and positive-generator inverse is encoded in the original triangle table. Let $\ell_* \geq 2$ be the maximum relator length. The statement below is not asserted with unchanged constants for a reference-pruned or triangle-pruned table.

Theorem I.1. *If $e_0 \in (0, 1]$ satisfies Eq. (C.8), then*

$$\Delta_J(\Phi_{\text{ref}}) \geq \frac{e_0^2}{4096\ell_*^2 d}, \quad \Delta_\diamond(\Phi_{\text{ref}}) \geq \frac{e_0^2}{4096\ell_*^2}. \quad (\text{I.1})$$

For $d = 44086$ and $\ell_* = 19$, the denominators are 65 188 028 416 and 1 478 656, respectively.

We prove both estimates, keeping their different input tests explicit. Let U implement an arbitrary finite matrix-tracial comparison Ψ and put $B_g = U_{p_g, p_g}$. For normalized Choi error $\delta = d_J(\Phi_{\text{ref}}, \Psi)$ set $k = \sqrt{d\delta}$. For half-diamond error $\eta = d_\diamond(\Phi_{\text{ref}}, \Psi)$ set $k = \sqrt{\eta}$. All bath blocks of U are contractions.

I.1 Singletons and individual entries

In the Choi case, the projection onto entries outside the singleton's row diagonal has zero target expectation. Its comparison expectation is at most δ , yielding

$$\tau(I - B_g B_g^*) \leq d\delta = k^2. \quad (\text{I.2})$$

In the half-diamond case, input $|p_g\rangle\langle p_g|$. The probability of leaving p_g is at most η , so

$$\tau(I - B_g^* B_g) \leq \eta = k^2. \quad (\text{I.3})$$

The two trace deficits agree. Complete the polar part of B_g to a unitary v_g in the same matrix algebra. For singular values $0 \leq s \leq 1$, $(1 - s)^2 \leq 1 - s^2$, hence

$$\|B_g - v_g\|_2 \leq k. \quad (\text{I.4})$$

Take a triangle position (a, b) carrying real coefficient $c \in \{3/5, 4/5, -3/5\}$ and label g . It is distinct from the singleton position. In the Choi case, the target annihilates the scalar vector $|a, b\rangle - c|p_g, p_g\rangle$. Testing its normalized rank-one projection gives

$$\|U_{ab} - cB_g\|_2 \leq \sqrt{1 + c^2} k.$$

For the half-diamond case, input the normalized entangled state

$$\frac{|b\rangle|0\rangle + |p_g\rangle|1\rangle}{\sqrt{2}}.$$

Its target output annihilates $|a\rangle|0\rangle - c|p_g\rangle|1\rangle$. The expectation of the corresponding normalized projector is $\|U_{ab} - cB_g\|_2^2 / [2(1 + c^2)]$ for the comparison and at most η . Thus

$$\|U_{ab} - cB_g\|_2 \leq \sqrt{2(1 + c^2)} k.$$

Together with Eq. (I.4), both cases give

$$\|U_{ab} - cv_g\|_2 < 3k. \quad (\text{I.5})$$

For the larger half-diamond constant this follows from $\sqrt{2(1 + c^2)} + |c| < 3$, checked directly for $|c| = 3/5, 4/5$.

I.2 Leakage in the row-orthogonality equation

Let $S = \{p, p + 1\}$ be the two system positions of a triangle. In the Choi case, a projection onto positions in one of these rows and outside S has zero target weight. The sum of squared bath 2-norms in that outside row is at most $d\delta = k^2$. Since a block row of a unitary has operator norm at most one, the product of the two outside rows has 2-norm at most k .

In the half-diamond case, each input basis state in S leaves S with probability at most η . Therefore

$$\sum_{a \notin S, b \in S} \|U_{ab}\|_2^2 \leq 2\eta.$$

Unitarity of U (each row and each column of bath blocks has total squared 2-norm one) gives equality of the total incoming and outgoing leakage sums:

$$\begin{aligned} \sum_{a \in S, b \notin S} \|U_{ab}\|_2^2 &= |S| - \sum_{a, b \in S} \|U_{ab}\|_2^2 \\ &= \sum_{a \notin S, b \in S} \|U_{ab}\|_2^2. \end{aligned}$$

Each outside row consequently has squared 2-norm at most 2η . The product of outside rows has 2-norm at most $\sqrt{2} k$. Rectangular Hilbert–Schmidt norms here use the original bath dimension as denominator.

The full row product is zero. Replacing the four within-block entries by their scalar multiples of v_g , using Eq. (I.5), costs at most $12k$. With $a = 3/5, b = 4/5$, this gives

$$ab \|v_0 v_x^* - v_y v_z^*\|_2 \leq \begin{cases} 13k, & \text{Choi,} \\ (12 + \sqrt{2})k, & \text{half-diamond.} \end{cases}$$

Since $ab = 12/25$, both cases imply, for $w_g = v_0^* v_g$,

$$\|w_x w_y - w_z\|_2 < 32k. \quad (\text{I.6})$$

For example, replacing $\sqrt{2}$ by $3/2$ already makes the larger coefficient smaller than 32.

I.3 Relator error and the witness trace

A relator of length ℓ has $\ell - 1$ prefix merges. Equation (I.6) therefore contributes at most $32(\ell - 1)k$. A negative generator label differs from the inverse of its positive label by at most $32k$, using its protected inverse equation. Replacing every negative letter and setting $\varphi(s) = w_{\ell(s)}$ for positive generators yields

$$\max_{r \in \mathcal{R}} \|\varphi(r) - I\|_2 \leq 64\ell_* k. \quad (\text{I.7})$$

The unitary $\varphi(j)$ is exactly the label w_{85} because j is a named positive generator of the presentation.

The off-diagonal target entry between the identity and witness singletons is zero. A traceless Hermitian matrix of trace norm 2δ has operator norm at most δ . In the Choi case, its entry bound and Eq. (I.4) give

$$|\tau(w_{85})| \leq d\delta + 2\sqrt{d\delta} = k^2 + 2k. \quad (\text{I.8})$$

For the half-diamond case, use the input-reference state $(|p_0\rangle|0\rangle + |p_j\rangle|1\rangle)/\sqrt{2}$. Its comparison cross entry is $\tau(B_0 B_{85}^*)/2$, so the same operator-norm observation gives

$$|\tau(w_{85})| \leq 2\eta + 2\sqrt{\eta} = 2k^2 + 2k. \quad (\text{I.9})$$

If the relevant distance is no larger than the right side of Eq. (I.1), then $64\ell_* k \leq e_0$ and $k \leq 1/4$. But Eqs. (I.8)–(I.9) imply, respectively,

$$\|\varphi(j) - I\|_2^2 = 2 - 2\operatorname{Re} \tau(w_{85}) \geq \frac{7}{8} \quad \text{or} \quad \frac{3}{4},$$

both larger than $1/4$, contradicting Eq. (C.8). This excludes every finite matrix-tracial comparison at the stated distance. Taking infima and closure proves Theorem I.1.

The half-diamond estimate removes the system-dimension loss because its probes use only the relevant input positions. It does not by itself provide the same dimension-free bound in Choi distance. The norm comparison $d_\diamond \leq d d_J$ recovers exactly the $1/d$ factor in the Choi assertion.

J Why the tolerance is not supplied by a coordinate gap

For $m \geq 2$, define

$$z_m = \frac{m^2 - 1 + 2mi}{m^2 + 1}.$$

Assign one of the positive generators, for example generator 22, the matrix $\operatorname{diag}(1, z_m)$ and all other positive generators I_2 . Exact rational arithmetic gives

$$|z_m|^2 = 1, \quad |1 - z_m|^2 = \frac{4}{m^2 + 1}. \quad (\text{J.1})$$

Since every relator has length at most 19, telescoping its unitary factors gives defect at most $38/\sqrt{m^2 + 1}$.

For finite m , the tuple's commutant is the diagonal algebra. On the off-diagonal matrix unit E_{12} , the conjugation energy

$$\frac{1}{4 \cdot 43} \sum_s \|[U_s, E_{12}]\|_2^2$$

divided by $\|E_{12}\|_2^2$ is exactly

$$\kappa_m = \frac{1}{43(m^2 + 1)} \longrightarrow 0. \quad (\text{J.2})$$

Thus arbitrarily small presentation defects can coexist with a closing coordinate conjugation gap. The witness approaches the identity too, so the example does not challenge Proposition C.2. It rules out inserting an exact Kazhdan constant into this coordinate estimate without an additional lifting or repair argument.

The internality theorem of Ref. [13] gives no rates for the steps used here: its Proposition 2.2 chooses the Markov averaging times by a diagonal argument, and the later tolerances tend to zero without rates. The normalization theorem of Ref. [11] is likewise qualitative. Neither supplies a numerical tolerance for the 4 417 relators here. A global numerical certificate therefore still requires one of two additional results: an evaluated tolerance in Eq. (C.8), uniform over all dimensions, or a quantitative support-leakage theorem that handles arbitrary nearby finite-tracial channels. Tests up to a finite bath dimension cannot replace either result.

K Causal purity and the sharper memory bound

We use the causal service model of Ref. [23, Definition 2.3]. A service retains Q qubits in an initial state β_0 , with purity $P = Q - S(\beta_0)$. Between rounds it exports j_t qubits permanently and imports j_t qubits in a fixed joint state α_t independent of everything else. Put

$$c_t = j_t - S(\alpha_t), \quad C = \sum_t c_t, \quad J = \sum_t j_t.$$

The qubits within an imported register may be correlated. Every retained register counts, and there is no free reset, postselection, or measurement record. Each output is released before the next input arrives. Error ϵ is the complete-experiment half trace distance maximized over adaptive observers with quantum references. Entropies are in bits; $\ln$ denotes the natural logarithm.

K.1 Small entropy production gives a nearby tracial channel

Lemma K.1 (Local tracialization, Ref. [23, Lemma 5.4]). *For a unitary U on $\mathbb{C}^d \otimes \mathbb{C}^M$ and bath state ρ , let*

$$\Psi(X) = \text{Tr}_M[U(X \otimes \rho)U^*], \quad T(\rho) = \text{Tr}_d[U(\tau_d \otimes \rho)U^*].$$

Then $a = S(T(\rho)) - S(\rho) \geq 0$, and there is a finite mixture Θ of finite matrix-tracial channels with

$$d_J(\Psi, \Theta) \leq \sqrt{2 \ln 2 a}. \quad (\text{K.1})$$

Proof. Diagonalize input and output bath states with decreasing eigenvalues λ_i and μ_j . If $U_{ji} \in M_d$ are the corresponding bath-indexed blocks, then $t_{ji} = d^{-1} \text{Tr}(U_{ji}^* U_{ji})$ is doubly stochastic and $\mu_j = \sum_i t_{ji} \lambda_i$. For probability distributions $p_{ji} = t_{ji} \lambda_i$ and $q_{ji} = t_{ji} \mu_j$, the relative entropy in bits is $D(p||q) = S(\mu) - S(\lambda) = a$. In particular $a \geq 0$, and Pinsker's inequality gives

$$T_0 := \sum_{ji} t_{ji} |\lambda_i - \mu_j| \leq \sqrt{2 \ln 2 a}.$$

Pairing the ordered spectra minimizes the absolute-difference transport cost, so $\sum_i |\lambda_i - \mu_i| \leq T_0$.

With $\lambda_{M+1} = 0$ and P_k the top- k input bath projection, write

$$\rho = \sum_{k=1}^M w_k P_k / k, \quad w_k = k(\lambda_k - \lambda_{k+1}).$$

Let Q_k be the top- k output bath projection. The compression $A_k = (I \otimes Q_k)U(I \otimes P_k)$ is a contraction between spaces of equal dimension dk . Complete its polar factor to a unitary V_k , so $A_k = V_k H_k$, and put

$$W_{k,\pm} = V_k(H_k \pm i\sqrt{I - H_k^2}).$$

These are unitaries. Averaging their conjugation channels replaces $A_k X A_k^*$ by $A_k X A_k^* + C_k X C_k^*$, where $C_k = V_k \sqrt{I - H_k^2}$. For input bath P_k/k , the averaged channel and the original one share the part from A_k ; their remaining positive Choi matrices have equal trace $e_k = k^{-1} \sum_{i \leq k < j} t_{ji}$. Their half trace distance is at most e_k .

Mixing these finite-tracial channels with weights w_k gives Θ with

$$\begin{aligned} d_J(\Psi, \Theta) &\leq \sum_k w_k e_k = \sum_{i < j} t_{ji} (\lambda_i - \lambda_j) \\ &= \frac{1}{2} \sum_{ij} t_{ji} |\lambda_i - \lambda_j| \\ &\leq \frac{1}{2} \left(T_0 + \sum_j |\mu_j - \lambda_j| \right) \leq T_0. \end{aligned}$$

The equality with the half sum uses the vanishing signed sum supplied by double stochasticity. This proves Eq. (K.1). $\square$

K.2 The entropy ledger

Feed fresh maximally entangled input-reference pairs in every round and retain all outputs and references. Let ρ_t be the incoming memory marginal and U_t the round unitary. Its entropy production is

$$a_t = S(\text{Tr}_d[U_t(\tau_d \otimes \rho_t)U_t^*]) - S(\rho_t).$$

The round's output-reference marginal is unchanged after release, so the complete-experiment error places it within ϵ of $J(\Phi)$. Lemma K.1 and the triangle inequality imply

$$a_t \geq \frac{(\Delta_J(\Phi) - \epsilon)_+^2}{2 \ln 2}.$$

Exporting j_t qubits lowers memory entropy by at most j_t ; importing the independent register raises it by $j_t - c_t$. Thus $S(\rho_{t+1}) \geq S(\rho_t) + a_t - c_t$. The initial memory entropy is $Q - P$ and the final entropy is at most Q . Telescoping gives $\sum_t a_t \leq P + C$ and therefore

$$P + C \geq \frac{n(\Delta_J(\Phi) - \epsilon)_+^2}{2 \ln 2}. \quad (\text{K.2})$$

This proof allows memory correlations with earlier outputs; it uses only the incoming marginal and the independence of each fresh probe. There is no restriction on Q or J .

K.3 The flat-probe purity budget

We reproduce the sharper form from Ref. [24], Proposition 8.7, to make its use independent of access to that companion paper.

Lemma K.2 (Surprisal variance, Ref. [23, Lemma 5.9]; see also Ref. [24, Lemma 8.8]). *If a state β on q qubits has eigenvalues β_i and deficit $D = q - S(\beta)$, the random variable $-\log_2 \beta_i$, sampled with probability β_i , has variance at most $(q + 2/\ln 2)D$.*

Proof. For $x \geq 0$ put $\phi(x) = x \ln x - x + 1$. Then

$$x(\ln x)^2 \leq (2 + \ln_+ x)\phi(x).$$

To check this, set $x = e^t$. For $t \geq 0$, the difference is $e^t(t - 2) + t + 2$, which and whose first derivative vanish at zero and whose second derivative is $te^t \geq 0$. For $t \leq 0$, the difference is $2(te^t - e^t + 1) - t^2e^t$, whose derivative is $-t^2e^t \leq 0$ and whose value at zero is zero. The limit $x = 0$ is immediate.

Let $m = 2^q$ and $x_i = m\beta_i$. Their average is one, $x_i \leq m$, and $m^{-1} \sum_i \phi(x_i) = D \ln 2$. Variance is unchanged by the additive constant q and is at most the corresponding second moment. Therefore

$$\text{Var}(-\log_2 \beta_i) \leq \frac{1}{m(\ln 2)^2} \sum_i x_i (\ln x_i)^2 \leq \left(q + \frac{2}{\ln 2}\right) D.$$

□

Proposition K.3 (Flat-probe budget, Ref. [24, Proposition 8.7]). *Suppose a pure input-reference probe has output ω flat of rank r under Φ . Every causal service of n uses with $\epsilon \leq 1/16$ and $J \leq n \log_2 r/2$ satisfies*

$$P + C < 3Q + 6. \quad (\text{K.3})$$

Proof. Put $b = P + C$ and suppose $b \geq 3Q + 6$. Preload all independent imports and keep all exports on the device side of a mathematical dilation. The device then starts with $L = Q + J$ qubits in the product state $\sigma_0 = \beta_0 \otimes \bigotimes_f \alpha_f$. This reorganization does not change the observer's reduced state. Every factor has at most Q qubits.

If σ_i are the eigenvalues, define $Z = L + \log_2 \sigma_i$, sampled with probability σ_i . Then $\mathbb{E}Z = b$. Independence and Lemma K.2 give

$$\text{Var } Z \leq \left(Q + \frac{2}{\ln 2}\right) b.$$

Feed n copies of the given pure probe. Conditioned on an eigenvector of σ_0 , the observer's final state τ_i has rank at most 2^L , since it is a marginal of a pure state with an L -qubit device. For a real k , there are at most 2^{L-k} eigenvalues $\sigma_i \geq 2^{k-L}$. The support projection E of the corresponding τ_i therefore has rank at most 2^{2L-k} and $\text{Tr}(E\tau_O) \geq \Pr(Z \geq k)$. The ideal state $\omega^{\otimes n}$ is flat of rank r^n . Hence, using $2J \leq n \log_2 r$,

$$\Pr(Z \geq k) \leq \epsilon + 2^{2Q-k}. \quad (\text{K.4})$$

Set $u = b - 2Q - 3 \geq b/3 + 1 > 0$. The one-sided Chebyshev inequality gives $\Pr(Z \geq b - u) \geq u^2/(\text{Var } Z + u^2)$. Equation (K.4), at $k = b - u = 2Q + 3$, bounds the same probability by $1/16 + 1/8 = 3/16$. Thus $13u^2 \leq 3 \text{Var } Z$ and

$$b < \frac{27}{13} \left(Q + \frac{2}{\ln 2}\right) < 3Q + 6,$$

a contradiction. The final constant follows from $\ln 2 > 9/13$; for an exact check, the first three positive terms of $2 \sum_{k \geq 0} (1/3)^{2k+1}/(2k+1)$ already exceed $9/13$. □

K.4 Application to the original and compact channels

On the 11 230 singleton positions of Φ_{ref} , the channel is complete dephasing. Half of a maximally entangled state on these positions gives the output

$$\frac{1}{r} \sum_{g \in \mathcal{L}} |p_g p_g\rangle \langle p_g p_g|, \quad r = 11\,230.$$

This is flat of rank r . Its complement has minimal dimension r , so the maximal complementary entropy is $\log_2 r$.

Combining Eqs. (K.2) and (K.3) gives

$$Q \geq \frac{n(\Delta_J(\Phi_{\text{ref}}) - \epsilon)_+^2}{6 \ln 2} - 2 \quad \left(\epsilon \leq \frac{1}{16}, J \leq \frac{n}{2} \log_2 r \right). \quad (\text{K.5})$$

At $\epsilon \leq \Delta_J(\Phi_{\text{ref}})/2$, this becomes

$$Q \geq \frac{n\Delta_J(\Phi_{\text{ref}})^2}{24 \ln 2} - 2. \quad (\text{K.6})$$

With the earlier budget $P + C \leq 14Q + 2 \log_2(16/7)$ of Ref. [23, Theorem 5.8] in place of Proposition K.3, the same argument gives the leading coefficient $1/(112 \ln 2)$; Proposition K.3 makes it $14/3$ times larger, although the additive constant is worse. The rank-rate restriction is not asserted to be the optimal exchange rate for a channel outside K_d .

For Φ_2 and Φ_{15} , the equal superposition of the two reference positions is a pure probe with flat rank-two output. Thus Eq. (K.5) holds for either channel, with its own global Choi gap and the budget $J \leq n/2$. This does not restore the original rank-11 230 probe. No result here identifies the maximal complementary entropy of a reference-pruned channel.

The support-restricted bounds of Appendix H and the half-diamond bound of Appendix I cannot be inserted as Δ_J in Eq. (K.2) without the appropriate global Choi comparison.

L Verification package and data format

Only Python 3.10 or later and its standard library are needed for the finite-data checks. The LaTeX sources are separate from those executable requirements. All paths in this section are relative to the `verification` folder of the repository. From that folder, run

```
python verify_release.py
```

This runs the two retained baseline suites, the independent reduced-data checker, scalar and probe checks, deliberate-corruption controls in normal and optimized Python, and deterministic re-emission. The run writes its record to `reports/verification.json`. The baseline suites retain their own detailed records. The quicker command `python verify_release.py --quick` omits those baseline suites and is not the complete verification.

L.1 Where the objects are stored

File	Content
baseline/verify/data/qi218/presentation.json	The original literal presentation, generator names, relator families, and distinguished word.
baseline/verify/data/qi219/channel.json.gz	The original fully referenced channel, all label words and exact normal forms, generator labels, and 16 428 triples.
data/deletion_certificates.json	The 15 990 retained original indices and all 438 free-group certificates for deleted equations.
data/one_anchor_channel.json.gz	The reduced channel Φ_1 on M_{31981} .
data/two_anchor_channel.json.gz	The reduced channel Φ_2 on M_{31982} .
data/fifteen_qubits_channel.json.gz	The main channel Φ_{15} on M_{32768} , including all 786 added identity positions.
data/summary.json	Construction counts, canonical hashes, restricted-gap squares, and explicit null fields for numerical global gaps.
reports/finite_instances.json	Written by the full run: independently computed Choi histograms, checked dimensions, and the rational restricted bounds of Table H.1.
channel_api.py	Exact action on sparse inputs and matrix units; no dense channel matrix is allocated.

All system indices are zero-based. An entry `[row, column, numerator]` in `kraus[g]` means the entry `numerator/5` of A_g . The field `generator_labels` is a list of pairs from signed generator numbers to label numbers. The field `labels` retains the baseline word representatives and exact group values. The triangle list consists of ordered triples of label numbers. For the padded channel, `anchor_labels` lists the two original reference labels; the separate field `identity_padding_count` records the 786 added identity singletons.

For any matrix unit, the channel action is exactly

$$\Phi(E_{uv}) = \frac{1}{25} \sum_g \sum_{(a,u,n) \in \text{kraus}[g]} \sum_{(b,v,k) \in \text{kraus}[g]} nk E_{ab}. \quad (\text{L.1})$$

For example,

```
python channel_api.py --variant fifteen_qubits --matrix-unit 0 1
python channel_api.py --variant fifteen_qubits --matrix-unit 2 2
```

The first returns zero, the off-diagonal entry of the two-reference dephasing corner. The second evaluates a nontrivial triangle column.

L.2 Rebuilding and independent checks

The baseline builder recreates the presentation and original channel from the finite rules. The new emitter reads that exact table, produces the deletion certificates, and writes all three reduced instances. To reconstruct the complete chain in a new directory, run

```
python rebuild_all.py --out rebuilt
python verify_reduced.py --data rebuilt/data
```

The independent reduced-data checker imports no emitter code. It obtains the same matrices by removing old system slots and reindexing the surviving entries; it then appends the specified identity positions. It verifies every free-group certificate using its own stack reduction, checks the protected inverse equations and label coverage, and recomputes both normalization identities and the full Choi weight distribution.

Deliberate corruptions include a Kraus coefficient, a triangle, a certificate sign, a certificate dependency on a deleted equation, a witness reference, a padding count, the coefficient denominator, and a restricted bound relabeled as global. The runner requires the corresponding semantic rejection, not merely any nonzero exit code. These tests also run under `python -O`; none of the acceptance conditions is an optimizable assertion.

L.3 Provenance and proof scope

The baseline presentation has SHA-256

```
c17a57463b1d3c327f13dab537b2a61e9cba300625d2009a90202022ac65adfe
```

and the uncompressed canonical baseline channel has SHA-256

```
38150decf3d203e3eabbc5dee8d8a3f2d6390aa960c50f90fe4c87b804729b68
```

The corresponding hash of the fifteen-qubit channel is

```
1236ae6e7d587c738baac7c0e5815c90ece46ff8c9392fd0012614aae6e1aa1b
```

A hash binds bytes; it is not a proof of an algebraic identity. Canonical uncompressed bytes are compared because gzip encodings can vary across library versions.

The release includes written proofs, not a formalization in a proof assistant. Its finite checks certify the concrete presentation data, the quotient computations, the sparse channel identities, and the supplied free-group implications. They do not prove the external centralizer theorem, the normalization theorem, or the general analytical estimates. The analytical group and channel claims rest on the written proofs and the external theorems; finite-data checks do not change their status.

The file `analytical_status.json` records each main claim and its dependency. Both `global_numerical_gap` and `relator_tolerance_e0` are null.

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